

ST90 Hybrid Stepper Motor NEMA 34

SKU / Model Number: ST90-N34-12NM

Manufacturer: PrecisionStep Inc

Category: Motors | ETIM Code: EC001855

PRODUCT OVERVIEW

The ST90 is a 2-phase hybrid stepper motor (NEMA 34 frame) for high-torque positioning applications including CNC routers, laser cutters, and 3D printers. Features 1.8 degree step angle and low detent torque.

TECHNICAL SPECIFICATIONS

Step Angle	1.8 degrees (+/- 5%)
Holding Torque	12 Nm
Detent Torque	0.4 Nm
Steps per Revolution	200
Rotor Inertia	3400 g.cm ²
Number of Leads	4 (bipolar)
Motor Length	150 mm

ELECTRICAL RATINGS

Rated Current	6.0 A / phase
Phase Resistance	0.36 ohm
Phase Inductance	3.5 mH
Rated Voltage	2.16 V DC (per phase)
Recommended Driver	24-80 V DC chopper

DIMENSIONS & MECHANICAL

Frame Size	NEMA 34 (86 x 86 mm)
Body Length	150 mm
Shaft Diameter	14 mm
Shaft Length	37 mm
Weight	5.2 kg

OPERATING CONDITIONS

Ambient Temperature	-20 to 50 deg C
Insulation Class	Class B (130 deg C)
Insulation Resistance	> 100 MOhm at 500 V DC
Dielectric Strength	500 V AC / 1 min
IP Rating	IP40

MATERIALS OF CONSTRUCTION

Stator	Silicon Steel Laminations
Rotor	NdFeB permanent magnets
Shaft	SUS420J2 Stainless Steel
Housing	Cold Rolled Steel, black anodized
Bearings	Ball bearings (sealed)

PERFORMANCE DATA / SELECTION TABLE

Speed (RPM)	Torque (Nm)	Phase Current (A)	Driver Voltage (V)
0	12.0	6.0	---
200	10.5	6.0	24
500	7.8	6.0	48
1000	4.2	6.0	48
2000	1.8	6.0	80

CERTIFICATIONS & APPROVALS

CE | RoHS | REACH

Note: Specific certification documents and test reports are available upon request. All products are manufactured under ISO 9001:2015 quality management system.

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